OPTIMIZATION

-Optimization is the act of obtaining the best result under given circumstances/ it can be defined as the process of finding the conditions that give the maximum or minimum value of a function.

Goal:

to minimize the effort required

to maximize the desired benefit

AEROSPACE:
MINIMIZATION of weight of aircraft is of major concern.

CHEMICAL ENGINEERS : design and operate a process plant for an optimum rate of production

MECHANICAL ENGINEERS:
Design mechanical components for the purpose for achieving either **a MINIMUM** manufacturing cost or a **MAXIMUM** component life

PRODUCTION ENGINEERS:
Design optimum schedule of various machining operations to **MINIMIZE** the idol time of machines and the overall job completion time

CIVIL ENGINEERS:
Design buildings, bridges,dams,and other structures in order to achieve a **MINIMUM** overall cost or **MAXIMUM** safety or both.

ELECTRICAL ENGINEERS:
Design communication networks so as to achieve **MINIMUM** time for communication from one node to another.

MAXIMIZE or MINIMIZE *Objective function*

Subject to *Constraints*

A solution of the model :is feasible ,if it satisfies all the constraints

: is optimal if, in addition to being feasible ,it yields the best (max or min) value of objective function

OPTIMAL PROBLEM FORMULATION

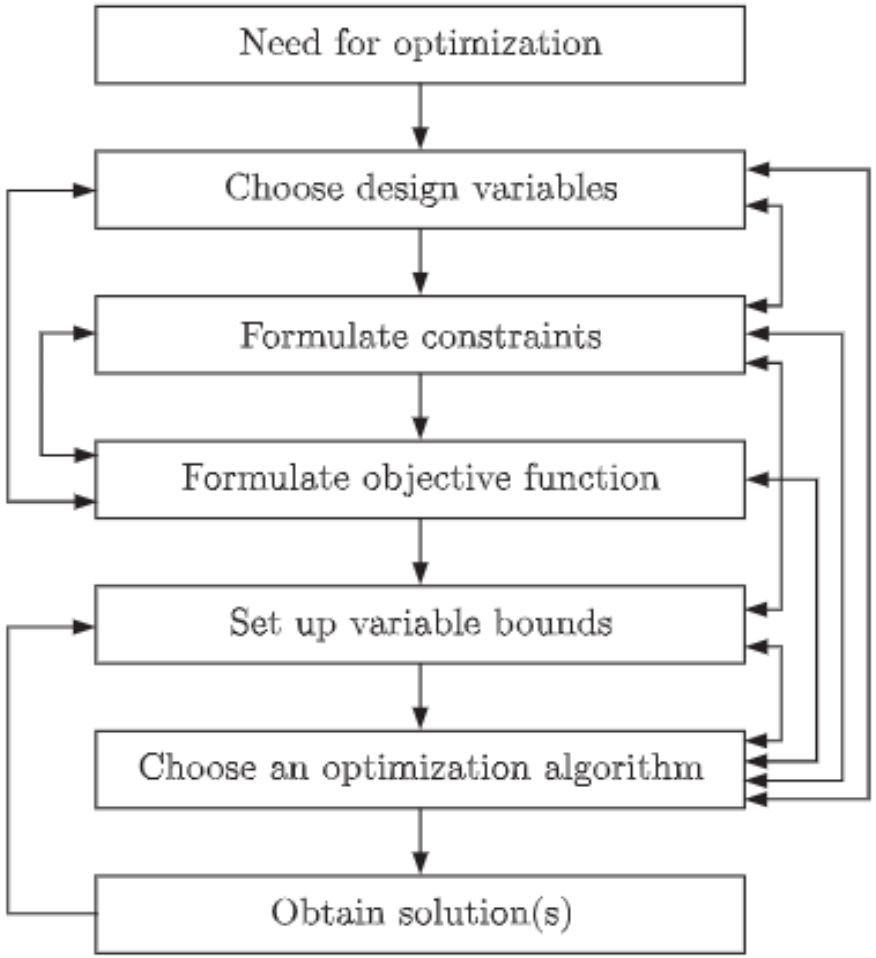


Figure 1.1 A flowchart of the optimal design procedure.

DESIGN VARIABLES

Design variables in optimization are **the independent, adjustable parameters (e.g., dimensions, material properties, shape) that an optimizer alters to minimize or maximize an objective function within specified constraints**.

- Design variables must be carefully chosen, as **a higher number of variables increases the complexity and computational cost of the optimization**. (first thumb rule of optimization: **choose few** design variables)

Key aspects of design variables include:

•**Definition:** They are the input quantities (like x1,x2,x3..xn) directly modified to improve performance, distinct from fixed parameters.

•**Types:** Variables can be geometric (lengths, angles), material properties, or operational settings.

•**Types of Values:** They may take scalar, vector, or matrix values.

•**Optimization Role:** They must be independent, and their range of allowed values constitutes the design constraints.

•**Examples:** In a structural beam optimization, design variables could be height, width, and material thickness.

•**Common Applications:** Used in shape, size, and topology optimization, such as determining material density (0 to 1) for components.

CONSTRAINTS

Two types of constraints
- inequality type
- equality type.

Inequality constraints state that the functional relationships among design variables are either greater than or equal to, or smaller than equal to, a resource value.

ex:

The stress ($^{\delta}(x)$) developed anywhere in a component must be smaller than or equal to the allowable strength ($S_{Allowable}$) of the material.

$$(^{\delta}(x)) \leq (S_{Allowable})$$

ex: The natural frequency (v(x)) of a system may required to be greater than 2 Hz,

$$(v(x)) \geq 2$$

Equality constraints state that the functional relationships should exactly match a resource value.

For example, a constraint may require that the deflection ($\delta(x)$) of a point in the component must be exactly equal to 5 mm, or mathematically,

$$\delta(x) = 5$$

CONSTRAINTS

- Equality constraints are usually more difficult to handle and, therefore, need to be avoided wherever possible.
- If the functional relationships of equality constraints are simpler, it may be possible to reduce the number of design variables by using the equality constraints. In such a case, the equality constraints reduce the complexity of the problem,thereby making it easier for the optimization algorithms to solve the problem.
- The number of complex equality constraints should be kept as low as possible

OBJECTIVE FUNCTION

Objective function:Maximized / Minimized

VARIABLE BOUNDS

- In general all N design variables are restricted to lie within minimum and maximum bounds as follows

$$x_i^{(L)} \leq x_i \leq x_i^{(U)} \quad \text{for } i = 1, 2, \ldots, N.$$

- Make a guess about the optimal solution and set the minimum and maximum bounds so that the optimal solution lies within these bounds
- After simulating the algorithm if the solution is found to lie within the chosen bounds, then it is ok.
- Otherwise if any design variables corresponding to the optimal solution is found to lie on or near the min/max bound, the choosen bound may be readjusted and simulation should be repeated
- Work only with linear problems

After the above four tasks are completed, the optimization problem can be mathematically written in a special format, known as *nonlinear programming* (NLP) format. Denoting the design variables as a column vector[1] $x = (x_1, x_2, \ldots, x_N)^T$, the objective function as a scalar quantity $f(x)$, J inequality constraints as $g_j(x) \geq 0$, and K equality constraints as

$$h_k(x) = 0$$

$$\text{Minimize} \quad f(x)$$

subject to

$$g_j(x) \geq 0, \qquad j = 1, 2, \ldots, J;$$

$$h_k(x) = 0, \qquad k = 1, 2, \ldots, K;$$

$$x_i^{(L)} \leq x_i \leq x_i^{(U)}, \qquad i = 1, 2, \ldots, N.$$

Classification of Optimization problems

Optimization problems can be classified based on

1. type of constraints
2. nature of design variables
3. physical structure of the problem
4. nature of the equations involved
5. deterministic nature of the variables
6. permissible value of the design variables
7. separability of the functions
8. number of objective functions.

Classification based on existence of constraints

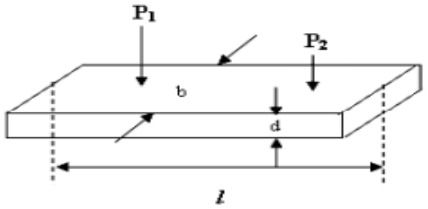
Constrained optimization problems: which are subject to one or more constraints.

Unconstrained optimization problems: in which no constraints exist

Classification based on the nature of the design variables

- the objective is to find a set of design parameters that make a prescribed function of these parameters minimum or maximum subject to certain constraints

For example to find the minimum weight design of a strip footing with two loads shown in the figure, subject to a limitation on the maximum settlement of the structure



The problem can be defined as follows

$$\text{Find } \mathbf{X} = \begin{Bmatrix} b \\ d \end{Bmatrix} \text{ which minimizes}$$

$$f(\mathbf{X}) = h(b,d)$$

subject to the constraints

$$\delta_s(\mathbf{X}) \le \delta_{max}$$

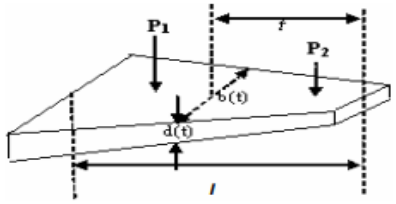
$$b \ge 0$$

$$d \ge 0$$

The length of the footing (l) the loads P1 and P2 , the distance between the loads are assumed to be constant and the required optimization is achieved by varying b and d. Such problems are called *parameter* or *static* optimization problems.

- the objective is to find a set of design parameters, which are all continuous functions of some other parameter, that minimizes an objective function subject to a set of constraints

For example, if the cross sectional dimensions of the rectangular footing is allowed to vary along its length as shown in the following figure



The problem can be defined as follows

$$\text{Find } \mathbf{X}(t) = \begin{Bmatrix} b(t) \\ d(t) \end{Bmatrix} \text{ which minimizes}$$

$$f(\mathbf{X}) = g(\, b(t), d(t)\,)$$

subject to the constraints

$$\delta_s(\mathbf{X}(t)) \le \delta_{max} \qquad 0 \le t \le l$$

$$b(t) \ge 0 \qquad 0 \le t \le l$$

$$d(t) \ge 0 \qquad 0 \le t \le l$$

The length of the footing (l) the loads P_1 and P_2 , the distance between the loads are assumed to be constant and the required optimization is achieved by varying b and d. Such problems are called *trajectory* or *dynamic* optimization problems.

Classification based on the physical structure of the problem

- **optimal control**
- **non-optimal control** problems.

(i) An optimal control (OC) problem is a mathematical programming problem involving a number of stages, where each stage evolves from the preceding stage in a prescribed manner.

It is defined by two types of variables: the control or design variables and state variables.

The problem is to find a set of control or design variables such that the total objective function (also known as the performance index, PI) over all stages is minimized subject to a set of constraints on the control and state variables.

(ii) The problems which are not optimal control problems are called non-optimal control problems.

Classification based on the nature of the equations involved

Based on the nature of expressions for the objective function and the constraints, optimization problems can be classified as

- **linear programming-**the objective function and all the constraints are linear functions of the design variables, the mathematical programming problem is called a linear programming (LP) problem
- **nonlinear programming-**any of the functions among the objectives and constraint functions is nonlinear, the problem is called a nonlinear programming (NLP) problem this is the most general form of a programming problem
- **Geometric-**geometric programming (GMP) problem is one in which the objective function and constraints are expressed as polynomials in X.
- **quadratic programming problems-**A quadratic programming problem is the best behaved nonlinear programming problem with a quadratic objective function and linear constraints and is concave (for maximization problems).

Classification based on the permissible values of the decision variable

(i) Integer programming problem -If some or all of the design variables of an optimization problem are restricted to take only integer (or discrete) values, the problem is called an integer programming problem.

(ii) Real-valued programming problem - A real-valued problem is that in which it is sought to minimize or maximize a real function by systematically choosing the values of real variables from within an allowed set. When the allowed set contains only real values, it is called a real-valued programming model

Classification based on deterministic nature of the variables

(i) Deterministic programming problem - all the design variables are deterministic.

(ii) Stochastic programming problem-some or all the parameters (design variables and/or pre-assigned parameters) are probabilistic (non deterministic or stochastic).

Classification based on separability of the functions

- In this type of a problem the objective function and the constraints are separable.
- A function is said to be separable if it can be expressed as the sum of n single-variable functions and separable programming problem can be expressed in standard form as :

Find **X** which minimizes $f(X) = \sum_{i=1}^{n} f_i(x_i)$

subject to : $g_j(X) = \sum_{i=1}^{n} g_{ij}(x_i) \le b_j \ , \quad j = 1,2,\ldots,m$

where b_j is a constant.

Classification based on the number of objective function

(i) Single-objective programming problem in which there is only a single objective.

(ii) Multi-objective programming problem-A multiobjective programming problem can be stated as follows:

Find $\mathbf{X}$ which minimizes $f_1(X), f_2(X), \ldots f_k(X)$

subject to : $g_j(\mathbf{X}) \leq 0, \qquad j = 1, 2, \ldots, m$

- where $f_1, f_2, \ldots f_k$ denote the objective functions to be minimized simultaneously.

Activate Wind

Example 1

- Wilson company produces three products: desk, chairs, and molded steel It is trying to decide on the number of desks (D), chair (C), and pound of model steels (M)

1. If the nets $ 50 profit on each desk produced, $30 on each chair produced, and $ 6 per pound of modeled steel produced. What the total profit?

50D+30C+6M

2. If 7 pounds of raw steel are needed to manufacture a desk, 3 pounds to manufacture a chair, and 1.5 pounds to produce a pound of model steel. What is the amount of raw steel used during the production?

$$7D + 3C + 1.5\,M$$

3- Wilson has only 2000 pounds of raw steel available and at least 100 desk must be produced to satisfy contract commitment.

$$7D + 3C + 1.5M <= 2000 \text{ (raw steel)}$$

$$D >= 100 \text{ (contract)}$$

MAXIMIZE $50D + 30C + 6M$

SUBJECT TO

$7D + 3C + 1.5M <= 2000$ (Raw steel)

$D >= 100$ (contract)

A furniture manufacturing company plans to make two products : chairs and tables. From its available resources which consists of 400 square feet to teak wood and 450 man hours. It is known that to make a chair requires 5 square feet of wood and 10 man-hours and yields a profit of Rs 45, while each table uses 20 square feet of wood and 25 man-hours and yields a profit of Rs 80. How many items of each product should be produced by the company so that the profit is maximum?

Let x units of chairs and y units of tables were produced Therefore $x, y \geq 0$

The given information can be tabulated as follows:

	Wood(square feet)	Man hours
Chairs(x)	5	10
Tables(y)	20	25
Availability	400	450

Therefore, the constraints are

$5x + 20y \leq 400$

$10x + 25y \leq 450$

It is known that to make a chair requires 5 square feet of wood and 10 man-hours and yields a profit of Rs 45, while each table uses 20 square feet of wood and 25 man-hours and yields a profit of Rs 80.

Therefore, profit gained to make x chairs and y tables is Rs $45x$ and Rs $80y$ respectively.

Total profit = Z = $45x + 80y$

which is to be maximised.

$$\text{Max } Z = 45x + 80y$$

subject to

$$5x + 20y \leq 400$$

$$10x + 25y \leq 450$$

$$x, y \geq 0$$

Product Mix Decision @ Whoppy Soft Drinks

The production manager for the Whoppy soft drink company is considering the production of 2 kinds of soft drinks: regular (R) and diet (D). The company operates on “8 hour” shift per day. Therefore, the production time is 480 minutes per day. During the production process, one of the main ingredients, syrup is limited to maximum production capacity of 675 gallons per day. Production of a regular case requires 2 minutes and 5 gallons of syrup, while production of a diet case needs 4 minutes and 3 gallons of syrup. Profits for regular soft drink are Rs.3.00 per case and profits for diet soft drink are Rs.2.00 per case. Write the formulation for this linear program.

Let R = number of regular drinks produced per days
D = number of diet drinks produced per days

Therefore, the objective is to maximize the total contribution from these two products.

Max Z= 3R + 2D

This contribution realization is subject to the following constraints

$2R + 4D \leq 480$ ---------- (Production time constraint)
$5R + 3D \leq 675$ ---------- (Syrup availability constraint)

R & D cannot be negative quantities,
hence,
R>=0 & D>=0

Sales Mix Decision for computer retail sales

A computer retail store sells two types of flat screen monitors: 17 inches and 19 inches, with a profit contribution of Rs. 300 and Rs. 250, respectively. The monitors are ordered each week from an outside supplier. As an added feature, the retail store installs on each monitor a privacy filter that narrows the viewing angle so that only persons sitting directly in front of the monitor are able to see on-screen data. Each 19” monitor consumes about 30 minutes of installation time, while each 17” monitor requires about 10 minutes of installation time. The retail store has approximately 40 hours of labor time available each week. The total combined demand for both monitors is at least 40 monitors each week. How many units of each monitor should the retail store order each week to maximize its weekly profits and meet its weekly demand?

Let $X1$ = number of 17 inches monitor to be ordered per week
$X2$ = number of 19 inches monitor to be ordered per week

Therefore, the objective of the retail service firm is to maximize the total contribution from these computer monitor sales.
The retail firm gets Rs. 300 and Rs. 250 per monitor for 17 and 19 inches respectively, therefore, the objective function arrived as follows:

$$\text{Max } Z = 300\,X1 + 250\,X2$$

This contribution realization is subject to the following constraints;

$10X1 + 30X2 \leq 2400$ ---------- (Labor time constraint)

$X1 + X2 \geq 40$ ---------- (Market demand for the computer monitor per week constraint)

And obviously, $X1$ & $X2$ cannot be negative quantities, hence

$X1 >= 0$ & $X2 >= 0$

Reddy Mikks produces both interior and exterior paints from two raw materials, Ml and *M2*. The following table provides the basic data of the problem:

	Tons of raw material per ton of		Maximum daily availability (tons)
	Exterior paint	*Interior paint*	
Raw material, *M*1	6	4	24
Raw material, *M*2	1	2	6
Profit per ton ($1000)	5	4	

A market survey indicates that the daily demand for interior paint cannot exceed that for exterior paint by more than 1 ton. Also, the maximum daily demand for interior paint is 2 tons.

Maximize $z = 5x_1 + 4x_2$

subject to

$6x_I + 4x_2 <= 24$ (1)

$x_I + 2x_2 =< 6$ (2)

$-x_1 + x_2 <= 1$ (3)

$X_2 <= 2$ (4)

$x_1, x_2 >= 0$ (5)

GRAPHICAL LP SOLUTION

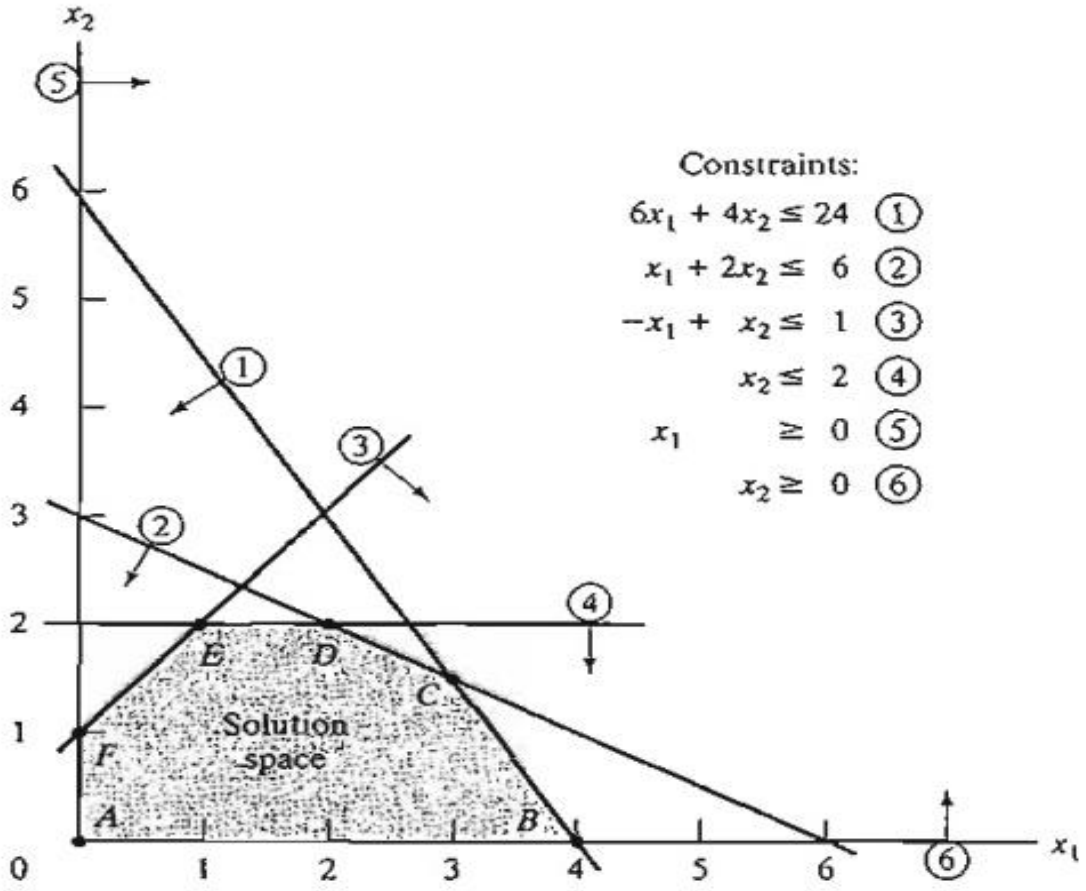
- Determination of the feasible solution space
- Determination of the optimum solution among all the points in the solution space

GRAPHICAL LP SOLUTION

- Change the inequality to equality….objective function and constraints
- Plot the non-negativity constraints
- Find the ‘x’ and ‘y’ intercept of each constraint equation
- Plot the constraints on x-y plane by connecting x and y intercept
- Identify the feasible solution space
- Find all the inner points of the feasible solution space
- Among the corner points select the one that optimize the objective function(optimal solution)

Solution of Maximization model

Feasible space of the Reddy Mikks model



Solution of Maximization model

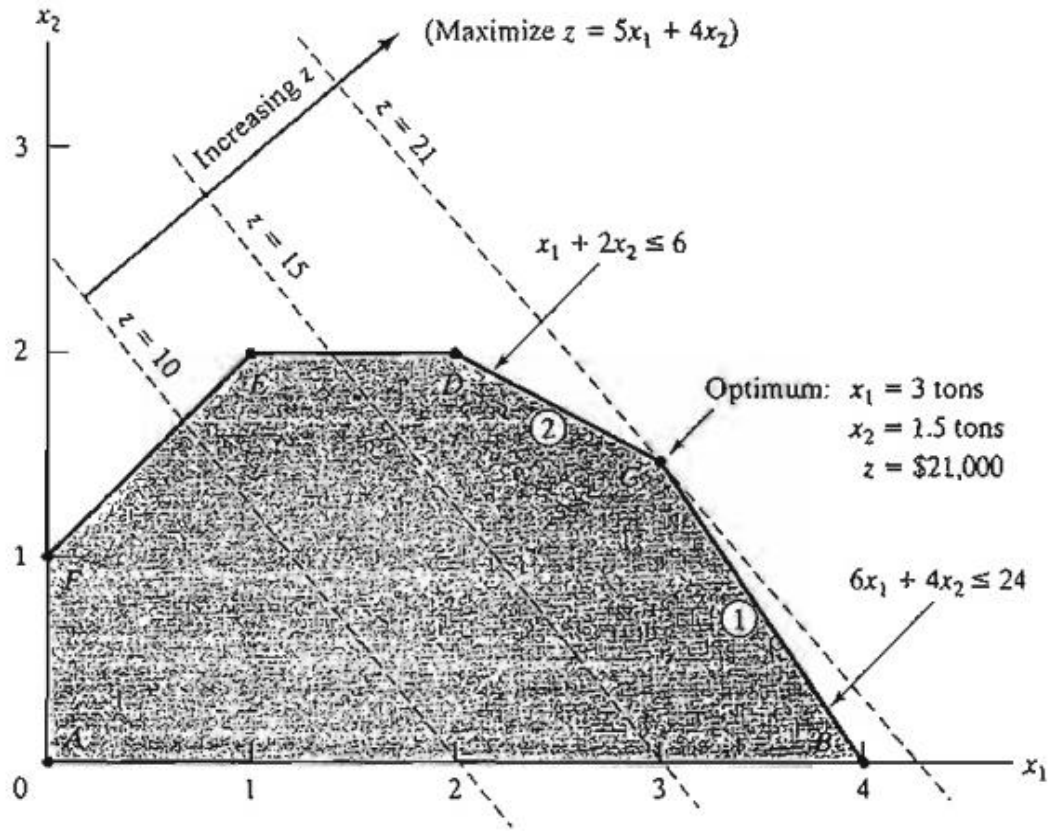


FIGURE 2.2

Optimum solution of the Reddy Mikks model

Example: Graphic Solution of LP Model

1. Find the optimum solution for the following LP using graphical method

Maximization: $Z = 60X+50Y$

Subjected to:

$4X + 10Y \leq 100$ hours → Assembly time

$2X + Y \leq 22$ hours → Inspection time

$3X + 3Y \leq 39\ m^3$ → Storage space

$X, Y \geq 0$

y = -2x + 22
y = -0.4x + 10
y = -x + 13
25
20
15
10
5
0
0
5
10
15
20
25
30
1
2
3
4
5

Classical optimization techniques

- useful in finding the optimum solution or unconstrained maxima or minima of continuous and differentiable functions.
- These are analytical methods and make use of differential calculus in locating the optimum solution.
- These methods assume that the function is differentiable twice with respect to the design variables and that the derivatives are continuous.
- Three main types of problems can be handled by the classical optimization techniques, viz.,

  single variable functions,

  multivariable functions with no constraints and

  multivariable functions with both equality and inequality constraints.

Assignment 1

- Optimality conditions,convexity and unimodality-[define how to identify optimal solutions] dtd 7/3

Linear Programming

- Linear Programming (LP) is the most useful optimization technique used for the solution of engineering problems.
- The term ‘linear’ implies that the objective function and constraints are ‘linear’ functions of

‘non-negative’ decision variables.

- The conditions of LP problems (LPP) are

1. Objective function must be a linear function of decision variables
2. Constraints should be linear function of decision variables
3. All the decision variables must be nonnegative

General form

Maximize	$Z = 6x + 5y$	π Objective Function
subject to	$2x - 3y \le 5$	π 1st Constraint
	$x + 3y \le 11$	π 2nd Constraint
	$4x + y \le 15$	π 3rd Constraint
	$x,\ y \ge 0$	π Nonnegativity Condition

Standard form of LPP

Standard form of LPP must have following three characteristics:

1. Objective function should be of maximization type
2. All the constraints should of equality type
3. All the decision variables should be nonnegative

Standard form of LPP-Conversion

Let us consider the following example.

$$\begin{aligned} \text{Minimize} \quad & Z = -3x_1 - 5x_2 \\ \text{subject to} \quad & 2x_1 - 3x_2 \le 15 \\ & x_1 + x_2 \le 3 \\ & 4x_1 + x_2 \ge 2 \\ & x_1 \ge 0 \\ & x_2 \text{ unrestricted} \end{aligned}$$

The above LPP is violating the following criteria of standard form:

1. Objective function is of minimization type
2. Constraints are of inequality type
3. Decision variable x_2 is unrestricted, i.e., it can take negative values also, thus violating the non-negativity criterion.

- Objective function is rewritten as {multiply by -1}

$$\text{Maximize} \quad Z' = -Z = 3x_1 + 5x_2$$

Converting inequalities into equations with non-negative RHS

- To convert '≤' to '=' a non-negative slack variable is added to LHS

$$2x_1 - 3x_2 + x_3 = 15.$$

x_3 is the slack variable (or unused amount)of the constraint

- To convert '≥' to '=' is achieved by subtracting a non-negative surplus variable from LHS side of inequality.

$$4x_1 + x_2 - x_5 = 2$$

x_5 surplus variable

Dealing with unrestricted variables

Decision variable x_2 can expressed by introducing two extra nonnegative variables as

$$x_2 = x_2' - x_2''$$

Thus, x_2 can be negative if $x_2' < x_2''$ and positive if $x_2' > x_2''$ depending on the values of x_2' and x_2''. x_2 can be zero also if $x_2' = x_2''$.

standard form of above LPP is as follows:

$$\text{Maximize} \quad Z' = -Z = 3x_1 + 5\left(x_2' - x_2''\right)$$

$$\text{subject to} \quad 2x_1 - 3\left(x_2' - x_2''\right) + x_3 = 15$$

$$x_1 + \left(x_2' - x_2''\right) + x_4 = 3$$

$$4x_1 + \left(x_2' - x_2''\right) - x_5 = 2$$

$$x_1,\ x_2',\ x_2'',\ x_3,\ x_4,\ x_5 \geq 0$$

Simplex method

The steps of the simplex method:

Step 1: Determine a starting basic feasible solution.

Step 2: Select an entering variable using the optimality condition. Stop if there is no entering variable.

Step 3: Select a leaving variable using the feasibility condition.

Optimality condition:

The entering variable in a maximization (minimization) problem is the non-basic variable having the most negative (positive) coefficient in the Z-row. The optimum is reached at the iteration where all the Z-row coefficient of the non-basic variables are non-negative (non-positive).

Feasibility condition: For both maximization and minimization problems the leaving variable is the basic associated with the smallest non-negative ratio (with strictly positive denominator)

Simplex method

1.Pivot row:

a) Replace the leaving variable in the basic column with the entering variable.

b) New pivot row = current pivot row / pivot element.

2. All other rows: New row=current row - (pivot column coefficient) *new pivot row.

Example 1:

Use the simplex method to solve the (LP) model:

$$max\ Z = 5x_1 + 4x_2$$

Subject to

$$6x_1 + 4x_2 \leq 24$$

$$x_1 + 2x_2 \leq 6$$

$$-x_1 + x_2 \leq 1$$

$$x_2 \leq 2$$

$$x_1, x_2 \geq 0$$

$$\text{Maximize } z = 5x_1 + 4x_2 + 0s_1 + 0s_2 + 0s_3 + 0s_4$$

subject to

$$\begin{aligned}
6x_1 + 4x_2 + s_1 \qquad\qquad\qquad &= 24 \\
x_1 + 2x_2 \qquad + s_2 \qquad\qquad &= 6 \\
-x_1 + x_2 \qquad\qquad + s_3 \qquad &= 1 \\
x_2 \qquad\qquad\qquad\qquad + s_4 &= 2
\end{aligned}$$

$$x_1,\ x_2,\ s_1,\ s_2,\ s_3,\ s_4 \geq 0,$$

where the variables s_1, s_2, s_3, s_4 are slack variables.

Solution. We write the objective equation as

$$z - 5x_1 - 4x_2 = 0.$$

m

no. of linear equations

n

no. of non-negative variables

m<n

(n-m)

M=4

n=6 (n-m) non-basic variables are

Basic	z	x_1	x_2	s_1	s_2	s_3	s_4	Solution	
z	1	−5	−4	0	0	0	0	0	z-row
s_1	0	6	4	1	0	0	0	24	s_1-row
s_2	0	1	2	0	1	0	0	6	s_2-row
s_3	0	−1	1	0	0	1	0	1	s_3-row
s_4	0	0	1	0	0	0	1	2	s_4-row

In our example, x_1 has the most negative coefficient -5 and hence x_1 is the entering variable. Now, we compute the ratios of $\frac{\text{Solution}}{\text{coefficient of } x_1}$ for "s_1-row, s_2-row, s_3-row and s_4-row". The table of ratios is given below.

Basic	**Entering** x_1	**Solution**	**Ratio (or intercept)**
s_1	6	24	$x_1 = \frac{24}{6} = 4 \leftarrow$ minimum
s_2	1	6	$x_1 = \frac{6}{1} = 6$
s_3	-1	1	$x_1 = \frac{1}{-1} = -1$ (negative denominator, ignore)
s_4	0	2	$x_1 = \frac{2}{0}$ (zero denominator, ignore)

			Enter ↓							
	Basic	z	x_1	x_2	s_1	s_2	s_3	s_4	**Solution**	
	z	1	−5	−4	0	0	0	0	0	
Leave ←	s_1	0	6	4	1	0	0	0	24	Pivot row
	s_2	0	1	2	0	1	0	0	6	
	s_3	0	−1	1	0	0	1	0	1	
	s_4	0	0	1	0	0	0	1	2	
			Pivot column							

Next, we compute the **Gauss-Jordan row operations** to obtain the new basic solutions. These computations are carried out in the following two steps:

1. For pivot row:
   (a) Replace the leaving variable in the "Basic" column by the entering variable.
   (b) New pivot row = Current pivot row ÷ Pivot element.
2. For all other rows:

$$\text{New row} = \big(\text{Current row}\big) - \big((\text{Pivot column coefficient}) \times (\text{New pivot row})\big).$$

These computations are shown below:

1. Replace s_1 by x_1 in the Basic column.

$$\begin{aligned}\text{New } x_1-\text{row} &= \text{Current } s_1\text{-row} \div 6\\ &= \frac{1}{6}\begin{pmatrix}0 & 6 & 4 & 1 & 0 & 0 & 0 & 24\end{pmatrix}\\ &= \begin{pmatrix}0 & 1 & \frac{2}{3} & \frac{1}{6} & 0 & 0 & 0 & 4\end{pmatrix}.\end{aligned}$$

2. $$\begin{aligned}\text{New } z-\text{row} &= \text{Current } z\text{-row} - (-5)\times \text{New } x_1 \text{ row}\\ &= \begin{pmatrix}1 & -5 & -4 & 0 & 0 & 0 & 0 & 0\end{pmatrix} - (-5)\begin{pmatrix}0 & 1 & \frac{2}{3} & \frac{1}{6} & 0 & 0 & 0 & 4\end{pmatrix}\\ &= \begin{pmatrix}1 & 0 & -\frac{2}{3} & \frac{5}{6} & 0 & 0 & 0 & 20\end{pmatrix}.\end{aligned}$$

3. $$\begin{aligned}\text{New } s_2-\text{row} &= \text{Current } s_2\text{-row} - (1)\times \text{New } x_1 \text{ row}\\ &= \begin{pmatrix}0 & 1 & 2 & 0 & 1 & 0 & 0 & 6\end{pmatrix} - (1)\begin{pmatrix}0 & 1 & \frac{2}{3} & \frac{1}{6} & 0 & 0 & 0 & 4\end{pmatrix}\\ &= \begin{pmatrix}0 & 0 & \frac{4}{3} & -\frac{1}{6} & 1 & 0 & 0 & 2\end{pmatrix}.\end{aligned}$$

4. New s_3 – row $=$ Current s_3-row $-(-1)\times$ New x_1 row

$$= \begin{pmatrix} 0 & -1 & 1 & 0 & 0 & 1 & 0 & 1 \end{pmatrix} - (-1)\begin{pmatrix} 0 & 1 & \frac{2}{3} & \frac{1}{6} & 0 & 0 & 0 & 4 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & \frac{5}{3} & \frac{1}{6} & 0 & 1 & 0 & 5 \end{pmatrix}.$$

5. New s_4 – row $=$ Current s_4-row $-(0)\times$ New x_1 row

$$= \begin{pmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 1 & 2 \end{pmatrix} - (0)\begin{pmatrix} 0 & 1 & \frac{2}{3} & \frac{1}{6} & 0 & 0 & 0 & 4 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 1 & 2 \end{pmatrix}.$$

				$\downarrow$					
	Basic	z	x_1	x_2	s_1	s_2	s_3	s_4	**Solution**
	z	1	0	$-\frac{2}{3}$	$\frac{5}{6}$	0	0	0	20
	x_1	0	1	$\frac{2}{3}$	$\frac{1}{6}$	0	0	0	4
$\leftarrow$	s_2	0	0	$\frac{4}{3}$	$-\frac{1}{6}$	1	0	0	2
	s_3	0	0	$\frac{5}{3}$	$\frac{1}{6}$	0	1	0	5
	s_4	0	0	1	0	0	0	1	2

Basic	Entering x_2	Solution	Ratio (or intercept)
x_1	$\frac{2}{3}$	4	$x_2 = 4 \div \frac{2}{3} = 6$
s_2	$\frac{4}{3}$	2	$x_2 = 2 \div \frac{4}{3} = 1.5$ (minimum)
s_3	$\frac{5}{3}$	5	$x_2 = 5 \div \frac{5}{3} = 3$
s_4	1	2	$x_2 = \frac{2}{1} = 2$

Basic	z	x_1	x_2	s_1	s_2	s_3	s_4	Solution
z	1	0	0	$\frac{3}{4}$	$\frac{1}{2}$	0	0	21
x_1	0	1	0	$\frac{1}{4}$	$-\frac{1}{2}$	0	0	3
x_2	0	0	1	$-\frac{1}{8}$	$\frac{3}{4}$	0	0	$\frac{3}{2}$
s_3	0	0	0	$\frac{3}{8}$	$-\frac{5}{4}$	1	0	$\frac{5}{2}$
s_4	0	0	0	$\frac{1}{8}$	$-\frac{3}{4}$	0	1	$\frac{1}{2}$

Observe that none of the coefficients of the z-row in the above table have negative coefficients. Hence the above table is optimal. The optimum is $z = 21$ and the optimal point is $(x_1, x_2) = \left(3, \frac{3}{2}\right)$. The complete basic solution is $(x_1, x_2, s_3, s_4) = \left(3, \frac{3}{2}, \frac{5}{2}, \frac{1}{2}\right)$. $\square$

Solve the following problem by simplex method.

$$\text{Max. } z = 4x_1 + 3x_2 + 6x_3$$

subject to

$$\begin{aligned} 2x_1 + 3x_2 + 2x_3 &\leq 440 \\ 4x_1 + 3x_3 &\leq 470 \\ 2x_1 + 5x_2 &\leq 430 \\ x_1, x_2, x_3 &\geq 0. \end{aligned}$$

Example 2:

Use the simplex method to solve the (LP) model:

$$max\ Z = 2x_1 + 3x_2$$

Subject to

$$0.25x_1 + 0.5x_2 \leq 40$$

$$0.4x_1 + 0.2x_2 \leq 40$$

$$0.8x_2 \leq 40$$

$$x_1, x_2 \geq 0$$